

Lagrangian Mechanics: Generalized Coordinates

Classical Mechanics – Undergraduate Notes

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1. Introduction and Degrees of Freedom

In the conventional Newtonian formulation, the instantaneous spatial position of a set of N point particles is parametrized using position vectors in Cartesian coordinates:

$$\mathbf{r}_i = (x_i, y_i, z_i), \quad i = 1, \dots, N. \quad (1)$$

To describe this unconstrained ensemble completely, $3N$ independent scalar coordinates are required. We therefore say that the system has $3N$ **degrees of freedom**.

Definition 1.1 (Degrees of Freedom and Reduction by Constraints)

If a physical system of N particles is subject to k independent holonomic constraints, each constraint reduces the number of required variables by one. Therefore, the total number of independent **degrees of freedom** s (which determines the dimension of configuration space) is:

$$s = 3N - k. \quad (2)$$

2. Constraints

A **constraint** in classical mechanics is any geometric, kinematic, or physical limitation imposed on a system that prevents its free motion in Cartesian coordinate space.

Holonomic vs. Nonholonomic Classification

Definition 2.1 (Holonomic Constraints)

A constraint is **holonomic** when it can be formulated as a finite algebraic equation involving only particle positions and time:

$$f_\alpha(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t) = 0, \quad \alpha = 1, \dots, k. \quad (3)$$

Example: The confinement of a particle to the surface of a fixed sphere of constant radius a , described by $x^2 + y^2 + z^2 - a^2 = 0$.

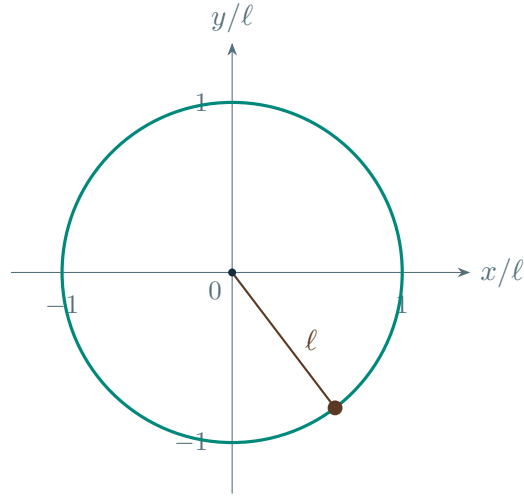


Figure 1: Planar example of a holonomic constraint: $x^2 + y^2 - \ell^2 = 0$, with fixed $\ell > 0$. The circle shows the admissible positions, and the segment indicates the radius ℓ . The axes are normalized by ℓ ; no time evolution is prescribed.

Nonholonomic Constraints

A constraint is **nonholonomic** if it cannot be reduced to integrable algebraic relations among position coordinates. These include:

- **Geometric inequalities:** Constraints of the form $f(\mathbf{r}_1, \dots, \mathbf{r}_N) \geq 0$ (e.g., a particle confined inside a spherical container of radius a , $r \leq a$).
- **Nonintegrable differential relations:** Kinematic constraints among coordinate and time differentials:

$$\sum_{i=1}^n A_{\alpha i} dq_i + B_{\alpha} dt = 0, \quad (4)$$

typical of pure rolling without slipping.

Scleronomous vs. Rheonomous Classification

- **Scleronomous Constraint:** The constraint does not depend explicitly on time:

$$f(\mathbf{r}_1, \dots, \mathbf{r}_N) = 0. \quad (5)$$

- **Rheonomous Constraint:** Time appears explicitly in its formulation:

$$f(\mathbf{r}_1, \dots, \mathbf{r}_N, t) = 0. \quad (6)$$

Example: A pendulum whose support oscillates vertically according to an imposed time law $z_0(t) = A \cos(\omega t)$.

3. Generalized Coordinates

Definition 3.1 (Generalized Coordinate)

A **generalized coordinate** (usually denoted by q_i , with $i = 1, \dots, s$) is any continuous parameter or variable used to specify uniquely the complete geometric configuration of the physical system.

- They are not restricted to being orthogonal components of a Euclidean vector or metric distances; they may represent angles, areas, relative lengths, or abstract quantities.
- They are assumed to be functionally and linearly independent, forming a **minimal set** of parameters that intrinsically satisfies the imposed holonomic constraints.

Example: For a particle on a sphere of radius a , instead of three algebraically constrained Cartesian coordinates (x, y, z) , two independent angular coordinates suffice: the colatitude θ and the longitude ϕ .

4. Point Transformations and the Jacobian Matrix

Definition 4.1 (Point Transformation)

A **point transformation** is a mapping or change of variables defined directly on configuration space. It relates a set of generalized coordinates $q = (q_1, \dots, q_s)$ to a new set $Q = (Q_1, \dots, Q_s)$:

$$Q_i = Q_i(q_1, q_2, \dots, q_s, t) \iff q_i = q_i(Q_1, Q_2, \dots, Q_s, t). \quad (7)$$

For this change of variables, the **Jacobian matrix** J contains the partial derivatives of the new coordinates with respect to the original ones:

$$J = \begin{pmatrix} \frac{\partial Q_1}{\partial q_1} & \frac{\partial Q_1}{\partial q_2} & \dots & \frac{\partial Q_1}{\partial q_s} \\ \frac{\partial Q_2}{\partial q_1} & \frac{\partial Q_2}{\partial q_2} & \dots & \frac{\partial Q_2}{\partial q_s} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial Q_s}{\partial q_1} & \frac{\partial Q_s}{\partial q_2} & \dots & \frac{\partial Q_s}{\partial q_s} \end{pmatrix}, \quad J_{ij} = \frac{\partial Q_i}{\partial q_j}. \quad (8)$$

Invertibility and Functional Independence

By the Inverse Function Theorem, for the transformation to be physically admissible and locally bijective (ensuring a well-defined inverse relation), the determinant of the Jacobian matrix must be nonzero:

$$\det(J) \neq 0. \quad (9)$$

This condition ensures that the new set of variables Q_i is functionally independent and does not collapse the dimensionality of the system.

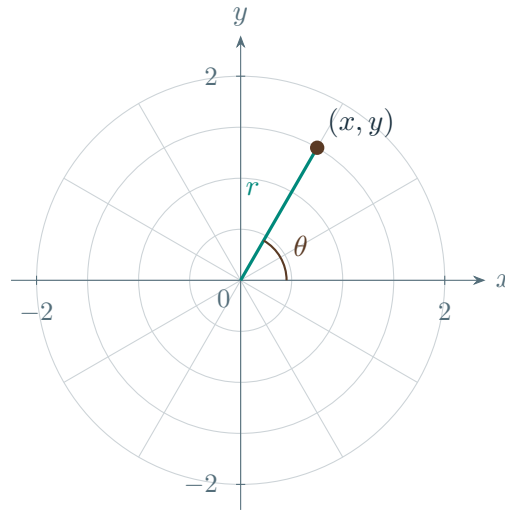


Figure 2: Polar transformation $x = r \cos \theta$, $y = r \sin \theta$. The circles and radial lines represent constant coordinates. The Jacobian satisfies $\det[\partial(x, y)/\partial(r, \theta)] = r$; a local coordinate chart requires $r > 0$ and an appropriate angular branch.

5. Kinematics: Generalized Velocities

Definition 5.1 (Generalized Velocity)

The total time derivative of a generalized coordinate q_i is called a **generalized velocity**:

$$\dot{q}_i \equiv \frac{dq_i}{dt}. \quad (10)$$

Unlike ordinary Cartesian velocity, its physical dimensions depend on the nature of q_i : if q_i is a linear distance, $\dot{q}_i$ has units of linear velocity (m/s); if q_i is an angle θ , then $\dot{q}_i = \dot{\theta}$ is an angular velocity (rad/s).

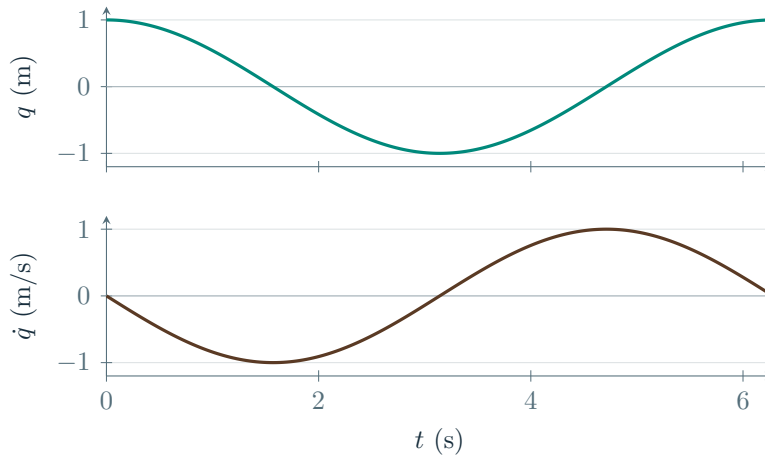


Figure 3: Position and velocity of an undamped harmonic oscillator: $q(t) = A \cos(\omega t)$ and $\dot{q}(t) = -A\omega \sin(\omega t)$. The values $A = 1 \text{ m}$ and $\omega = 1 \text{ s}^{-1}$ are used as illustrative parameters. The curves are analytical solutions, not measurements.

Relation to Cartesian Velocities

Given the functional relation $x_i = x_i(q_1, \dots, q_s, t)$, the Cartesian velocity $v_i = \dot{x}_i$ is obtained by applying the chain rule:

$$v_i = \sum_{k=1}^s \frac{\partial x_i}{\partial q_k} \dot{q}_k + \frac{\partial x_i}{\partial t}. \quad (11)$$

Fundamental Property: Dot-Cancellation Rule

When v_i is partially differentiated with respect to a specific generalized velocity $\dot{q}_j$, given that the functional dependence of v_i on the velocities $\{\dot{q}_k\}$ is strictly linear:

$$\frac{\partial v_i}{\partial \dot{q}_j} = \frac{\partial x_i}{\partial q_j}. \quad (12)$$

This identity from differential calculus is one of the essential tools in the formal derivation of the Euler–Lagrange equations from d’Alembert’s principle.

6. Configuration Space

Definition 6.1 (Configuration Space)

The **configuration space** is an abstract differentiable manifold $\mathcal{M}$ in which each point

uniquely represents the complete instantaneous geometric configuration of the physical system.

- **Coordinate axes:** They are defined by the set of independent generalized coordinates $(q_1, q_2, \dots, q_s)$.
- **Dimension:** The dimension of the manifold is exactly equal to the number of degrees of freedom: $\dim(\mathcal{M}) = s = 3N - k$.
- **Time evolution:** The system's dynamical history over time appears as a continuous curve or trajectory parametrized by t :

$$\mathbf{q}(t) = (q_1(t), q_2(t), \dots, q_s(t)) \in \mathcal{M}. \quad (13)$$

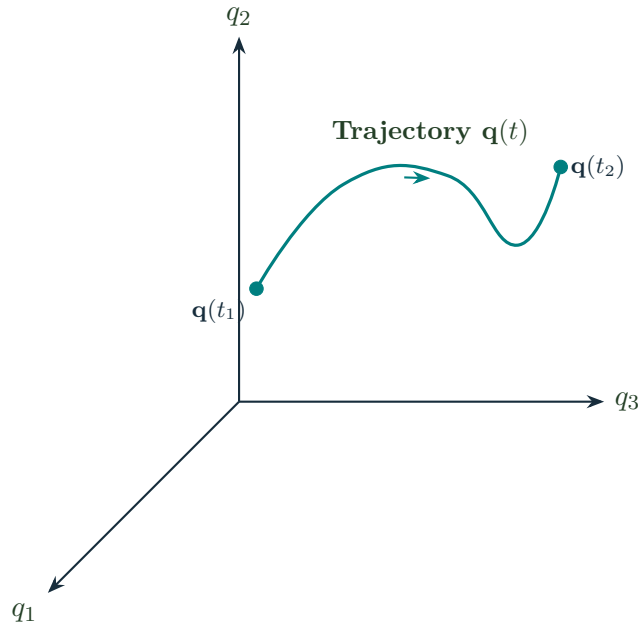


Figure 4: Curve or trajectory in the configuration space generated by the generalized coordinates. Time t acts as the parameter that traces the physical configuration of the system.